Understanding Variational Autoencoders from a mathematical view

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Feynman's words

What I cannot create, I do not understand.

Outline

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The Challenge of Generative Modeling

Given a dataset $\mathcal{D} = \{x^{(i)}\}_{i=1}^{N}$ sampled from an unknown distribution $p^*(x)$, how can we:

- Learn a model $p_\theta(x)$ that approximates $p^*(x)$?
- Generate new samples from the learned distribution?
- Perform inference: $p_\theta(z|x)$ for latent variables z?

Key challenges:

1. High-dimensional data spaces
2. Intractable likelihood computation
3. Latent variable inference
4. Scalable training procedures

Traditional Approaches and Their Limitations

- Maximum Likelihood Estimation (MLE):

$$\theta^* = \arg\max_{\theta} \sum_{i=1}^{N} \log p_\theta(x^{(i)}) \tag{1}$$

- Problem: Direct likelihood computation often intractable
- Latent Variable Models:

$$p_\theta(x) = \int p_\theta(x|z)p_\theta(z)dz \tag{2}$$

- Problem: Integration over latent space z is typically intractable

Solution: Variational inference to approximate intractable posteriors

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Variational Inference Framework

Given a latent variable model $p_\theta(x, z) = p_\theta(x|z)p_\theta(z)$, we want to:

- Approximate the intractable posterior $p_\theta(z|x)$
- Maximize the marginal likelihood $p_\theta(x)$

Key idea: Introduce a variational distribution $q_\phi(z|x)$ to approximate $p_\theta(z|x)$

$$\mathrm{KL}(q_\phi(z|x)\|p_\theta(z|x)) = \mathbb{E}_{q_\phi(z|x)}[\log q_\phi(z|x)] - \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(z|x)] \tag{3}$$

Evidence Lower Bound (ELBO)

The KL divergence can be decomposed as:

$$\mathrm{KL}(q_\phi(z|x)\|p_\theta(z|x)) = \log p_\theta(x) - \mathrm{ELBO}(x;\theta,\phi) \tag{4}$$

where the Evidence Lower Bound is:

$$\mathrm{ELBO}(x;\theta,\phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \mathrm{KL}(q_\phi(z|x)\|p_\theta(z)) \tag{5}$$

Key insight: Since $\mathrm{KL} \geq 0$, we have:

$$\log p_\theta(x) \geq \mathrm{ELBO}(x;\theta,\phi) \tag{6}$$

Maximizing ELBO $\Leftrightarrow$ Minimizing KL divergence $\Leftrightarrow$ Approximating posterior

ELBO Interpretation

The ELBO can be rewritten as:

$$\text{ELBO}(x; \theta, \phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \text{KL}(q_\phi(z|x) \| p_\theta(z)) \tag{7}$$

Two components:

1. Reconstruction term: $\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]$
   - Encourages $q_\phi(z|x)$ to encode information about x
   - Maximizes likelihood of reconstructing x from z
2. Regularization term: $-\text{KL}(q_\phi(z|x) \| p_\theta(z))$
   - Encourages $q_\phi(z|x)$ to be close to prior $p_\theta(z)$
   - Prevents overfitting to training data

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From Jensen to the ELBO

For any variational posterior $q_\phi(z \mid x)$,

$$\begin{aligned}
\log p_\theta(x) &= \log \int q_\phi(z \mid x)\, \frac{p_\theta(x,z)}{q_\phi(z \mid x)}\, dz \\
&\geq \mathbb{E}_{q_\phi}\left[\log p_\theta(x,z) - \log q_\phi(z \mid x)\right] \quad \text{(Jensen)} \\
&= \underbrace{\mathbb{E}_{q_\phi}\left[\log p_\theta(x \mid z)\right]}_{\text{reconstruction}} - \underbrace{\mathrm{KL}\left(q_\phi(z \mid x) \,\|\, p(z)\right)}_{\text{regularization}} \\
&=: \mathcal{L}(\theta, \phi; x) \quad \text{(ELBO)}.
\end{aligned}$$

Moreover, the identity

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + \mathrm{KL}\left(q_\phi(z \mid x) \,\|\, p_\theta(z \mid x)\right)$$

shows $\mathcal{L}$ is a lower bound and is tight iff $q_\phi(z \mid x) = p_\theta(z \mid x)$.

Two Equivalent Forms of the ELBO

(1) Recon - KL form

$$\mathcal{L} = \mathbb{E}_{q_\phi}\big[\log p_\theta(x \mid z)\big] - \mathrm{KL}\left(q_\phi(z \mid x) \,\|\, p(z)\right).$$

(2) Variational free energy

$$-\mathcal{L} = \underbrace{\mathrm{KL}\left(q_\phi(z \mid x) \,\|\, p_\theta(z \mid x)\right)}_{\text{inference gap}} - \log p_\theta(x).$$

At fixed θ, the $\mathcal{L}$-maximizer is the true posterior.

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VAE Architecture

Encoder-Decoder Architecture

- Encoder (Recognition model): $q_\phi(z|x)$
  - Maps data x to latent representation z
  - Typically: $q_\phi(z|x) = \mathcal{N}(z; \mu_\phi(x), \sigma^2_\phi(x)I)$
- Decoder (Generative model): $p_\theta(x|z)$
  - Reconstructs data x from latent z
  - Typically: $p_\theta(x|z) = \mathcal{N}(x; \mu_\theta(z), \sigma^2_\theta I)$ or Bernoulli
- Prior: $p_\theta(z) = \mathcal{N}(z; 0, I)$

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The Reparameterization Trick

Problem: How to backpropagate through stochastic sampling $z \sim q_\phi(z|x)$?
Solution: Reparameterization trick

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{8}$$

Benefits:

- Gradient flows through deterministic path: $x \to \mu_\phi(x), \sigma_\phi(x) \to z$
- Stochasticity moved to ϵ (independent of parameters)
- Enables end-to-end training with backpropagation

Alternative approaches:

- Score function estimator (higher variance)
- Control variates

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VAE Training Objective

For a dataset $\mathcal{D} = \{x^{(i)}\}_{i=1}^{N}$, the VAE loss is:

$$\mathcal{L}(\theta, \phi) = \sum_{i=1}^{N} \mathrm{ELBO}(x^{(i)}; \theta, \phi) \tag{9}$$

Expanding the ELBO:

$$\mathcal{L}(\theta, \phi) = \sum_{i=1}^{N} \left[\mathbb{E}_{q_\phi(z|x^{(i)})}[\log p_\theta(x^{(i)}|z)] - \mathrm{KL}(q_\phi(z|x^{(i)}) \| p_\theta(z)) \right] \tag{10}$$

Training procedure:

1. Sample $x^{(i)} \sim \mathcal{D}$
2. Sample $\epsilon \sim \mathcal{N}(0, I)$
3. Compute $z^{(i)} = \mu_\phi(x^{(i)}) + \sigma_\phi(x^{(i)}) \odot \epsilon$
4. Compute reconstruction loss: $\log p_\theta(x^{(i)}|z^{(i)})$
5. Compute KL divergence: $\mathrm{KL}(q_\phi(z|x^{(i)}) \| p_\theta(z))$
6. Backpropagate gradients

Analytical KL Divergence

For Gaussian distributions, the KL divergence has a closed form:

$$\mathrm{KL}(\mathcal{N}(\mu, \sigma^2) \| \mathcal{N}(0, I)) = \frac{1}{2} \sum_{j=1}^{J} (\mu_j^2 + \sigma_j^2 - \log \sigma_j^2 - 1) \tag{11}$$

where J is the dimension of the latent space.

Interpretation:

- μ_j^2: Penalizes large mean values
- $\sigma_j^2 - \log \sigma_j^2 - 1$: Encourages $\sigma_j \approx 1$
- Total effect: Regularizes latent space to be close to standard Gaussian

Benefits:

- No sampling required for KL term
- Stable gradients
- Interpretable regularization

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β-VAE: Controlling the Regularization

Motivation: Balance between reconstruction and regularization
β-VAE objective:

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \beta\, \mathrm{KL}(q_\phi(z|x) \| p_\theta(z)) \tag{12}$$

Effect of β:

- $\beta > 1$: Stronger regularization $\Rightarrow$ Better disentanglement
- $\beta < 1$: Weaker regularization $\Rightarrow$ Better reconstruction
- $\beta = 1$: Standard VAE

Disentanglement: β-VAE encourages learning of independent latent factors

VQ-VAE: Vector Quantized VAE

Key idea: Use discrete latent representations instead of continuous

Architecture:

- Encoder: $z_e = E(x)$ (continuous)
- Quantization: $z_q = \text{Quantize}(z_e)$ (discrete)
- Decoder: $\hat{x} = D(z_q)$

Quantization process:

$$z_q = \arg\min_k \|z_e - e_k\|^2 \tag{13}$$

where $\{e_k\}_{k=1}^K$ are learnable codebook vectors.

Benefits:

- Discrete representations (useful for language, etc.)
- Better compression
- Enables autoregressive modeling in latent space

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VQ-VAE Training

Challenge: Quantization is non-differentiable
Solution: Straight-through estimator

$$\frac{\partial \mathcal{L}}{\partial z_e} = \frac{\partial \mathcal{L}}{\partial z_q} \tag{14}$$

Training objective:

$$\mathcal{L} = \log p(x|z_q) + \| \mathrm{sg}(z_e) - e_k \|^2 + \beta \| z_e - \mathrm{sg}(e_k) \|^2 \tag{15}$$

where $\mathrm{sg}(\cdot)$ is stop-gradient operator.
Components:

- Reconstruction loss: $\log p(x|z_q)$
- Codebook loss: $\| \mathrm{sg}(z_e) - e_k \|^2$ (updates codebook)
- Commitment loss: $\beta \| z_e - \mathrm{sg}(e_k) \|^2$ (encourages encoder to commit)

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WAE: Wasserstein Autoencoder

Motivation: Replace KL divergence with Wasserstein distance
WAE objective:

$$\mathcal{L} = \mathbb{E}_{p(x)}\mathbb{E}_{q_\phi(z|x)}[c(x, D(z))] + \lambda\mathcal{D}(q_\phi(z), p(z)) \tag{16}$$

where:

- $c(x, D(z))$: Reconstruction cost
- $\mathcal{D}(q_\phi(z), p(z))$: Distance between aggregated posterior and prior
- λ: Regularization weight

Benefits:

- More flexible than KL divergence
- Better mode coverage
- Can use different distance metrics

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VAE-GAN: Combining VAE and GAN

Idea: Use GAN discriminator to improve VAE reconstruction quality

Architecture:

- VAE encoder-decoder: $x \to z \to \hat{x}$
- GAN discriminator: Distinguishes real vs reconstructed samples
- Generator: VAE decoder

Training objective:

$$\mathcal{L} = \mathcal{L}_{\text{VAE}} + \lambda \mathcal{L}_{\text{GAN}} \tag{17}$$

where:

- $\mathcal{L}_{\text{VAE}}$: Standard VAE loss
- $\mathcal{L}_{\text{GAN}}$: Adversarial loss for discriminator

Benefits:

- Better sample quality than VAE alone
- Maintains inference capabilities

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Hierarchical VAE

Motivation: Learn hierarchical latent representations

Architecture:

$$p_\theta(x, z_1, \ldots, z_L) = p_\theta(x|z_1) \prod_{l=1}^{L-1} p_\theta(z_l|z_{l+1}) p_\theta(z_L) \tag{18}$$

Inference:

$$q_\phi(z_{1:L}|x) = \prod_{l=1}^{L} q_\phi(z_l|z_{<l}, x) \tag{19}$$

Benefits:

- Multi-scale representations
- Better modeling of complex data
- Hierarchical generation process

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Conditional VAE (C-VAE)

Extension: Incorporate conditioning information

Model:

$$p_\theta(x, z|c) = p_\theta(x|z, c)p_\theta(z) \tag{20}$$

Inference:

$$q_\phi(z|x, c) = \mathcal{N}(z; \mu_\phi(x, c), \sigma_\phi^2(x, c)) \tag{21}$$

Applications:

- Conditional generation
- Controlled synthesis
- Multi-modal learning

Training:

$$\mathcal{L} = \mathbb{E}_{q_\phi(z|x,c)}[\log p_\theta(x|z, c)] - \mathrm{KL}(q_\phi(z|x, c)\|p_\theta(z)) \tag{22}$$

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Importance Weighted Autoencoder (IWAE)

Motivation: Better approximation of log-likelihood

IWAE objective:

$$\mathcal{L}_K = \mathbb{E}_{z_1,\ldots,z_K \sim q_\phi(z|x)} \left[\log \frac{1}{K} \sum_{k=1}^{K} \frac{p_\theta(x, z_k)}{q_\phi(z_k|x)} \right] \tag{23}$$

Key properties:

- $\mathcal{L}_K \geq \mathcal{L}_{K-1}$ (monotonic improvement)
- $\lim_{K\to\infty} \mathcal{L}_K = \log p_\theta(x)$ (exact likelihood)
- $K = 1$ recovers standard VAE

Trade-offs:

- Better approximation with larger K
- Higher computational cost
- More complex gradients

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VAE Implementation Details

Network architectures:

- Encoder: $x \to \mu_\phi(x), \log \sigma_\phi^2(x)$
- Decoder: $z \to \mu_\theta(z)$ (for Gaussian) or logits (for Bernoulli)
- Activation: ReLU, ELU, or Swish
- Normalization: BatchNorm or LayerNorm

Training tricks:

- KL annealing: Gradually increase KL weight
- Free bits: Prevent posterior collapse
- Warm-up: Start with reconstruction only

Common Issues and Solutions

Posterior collapse:

- Symptom: $q_\phi(z|x) \approx p_\theta(z)$ (no information in latent)
- Causes: Strong decoder, weak encoder
- Solutions: KL annealing, free bits, stronger encoder

Blurry reconstructions:

- Cause: Gaussian decoder assumption
- Solutions: VAE-GAN, better decoder architectures

Mode collapse:

- Symptom: Limited diversity in generated samples
- Solutions: Better regularization, architectural improvements

Evaluation Metrics

Reconstruction quality:

- MSE, SSIM, FID (Fréchet Inception Distance)
- Perceptual losses

Generation quality:

- FID, IS (Inception Score)
- Human evaluation

Representation quality:

- Disentanglement metrics (β-VAE metric, MIG)
- Downstream task performance
- Interpolation smoothness

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Recent Advances

Transformer-based VAEs:

- VQ-VAE-2: Hierarchical discrete representations
- DALL-E: Text-to-image generation
- VQGAN: High-quality image generation

Improved training:

- NVAE: Normalizing flow decoder
- VDVAE: Very deep VAE
- VDM: Diffusion-based decoder

Applications:

- Text generation (GPT-style models)
- Image synthesis and editing
- Audio generation
- Scientific modeling

Open Questions and Future Directions

Theoretical understanding:

- Why do VAEs work despite approximation errors?
- Optimal choice of prior distributions
- Connection to other generative models

Technical challenges:

- Better posterior approximations
- Scalable training for large models
- Integration with other architectures

Applications:

- Multi-modal learning
- Scientific discovery
- Efficient compression
- Robust representations

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Summary

Key contributions of VAE:

- Scalable variational inference for latent variable models
- End-to-end training with reparameterization trick
- Foundation for many generative models

VAE variants address:

- Disentanglement (β-VAE)
- Discrete representations (VQ-VAE)
- Sample quality (VAE-GAN, WAE)
- Hierarchical modeling (HVAE)

Impact:

- Influenced modern generative models
- Enabled practical latent variable modeling
- Foundation for many applications

Questions

If you are interested in the questions, please feel free to write an email to me (hpp@bimsa.cn) with your comments.

1. How does the choice of prior $p_\theta(z)$ affect VAE performance? What are the trade-offs between different priors?
2. Can you derive the gradient of the ELBO with respect to the encoder parameters ϕ? How does the reparameterization trick make this tractable?
3. What are the theoretical guarantees of VAE? Under what conditions does the ELBO provide a good approximation to the true log-likelihood?
4. How would you extend VAE to handle sequential data (like in language modeling)? What are the key challenges?

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Welcome inputs

Any comments?

Welcome your inputs!